



3 SIDES

SCALED

ISOSCELES

EQUILATERAL

OBTUSE

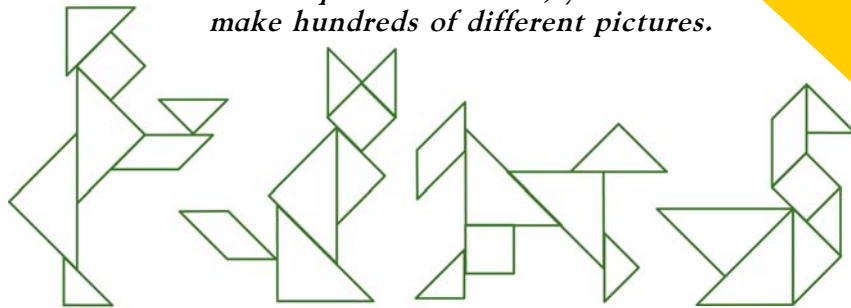
RIGHT-
ANGLED

Any shape made of straight lines can be split into triangles. Likewise, you can use triangles to create an endless variety of shapes. In China, people used this fact to invent a game called tangrams. The game reached the children of Europe and America about 100 years ago, when it caused a huge craze.

Triangles have special names depending on their sides and angles. If the sides are all equal, a triangle is *equilateral*. If the sides are all different, the triangle is *scalene*. If only two sides are the same, the triangle is *isosceles*. A triangle with one square side is *right-angled*, and one with an angle larger than 90° is *obtuse*.



Using just the 7 tangram pieces in the square on the left, you can make hundreds of different pictures.



Strong and simple

Triangles are the strongest of all simple shapes because, unlike squares and rectangles, their angles can't wobble. That's why you'll find triangles in bridges, buildings, and the girders of the Eiffel Tower in Paris.

How
do you
measure the
height of a tree
without climbing it?

Answer: use a right-angled triangle. As long as you know the size of one of the angles and one of the sides in a right-angled triangle, you can calculate the others. This branch of maths goes by the scary name of *trigonometry*.

